Plz introduce the study of Alzheimer's Disease of human. Plz first introduce the research background (when does it first studied and how). Then the progress from 1900 to 2000. Only select the most important findings.

👀 **Monica**

Here's a structured, human-focused introduction to Alzheimer's disease (AD) research—starting with how it was first studied, then tracing the **most influential discoveries from 1900–2000**. I'll keep it selective: the points below are the ones that most directly changed scientific understanding, diagnosis, or treatment direction.

🧭 Research background: how Alzheimer's disease first entered science

Alzheimer's disease research began as **careful clinical observation paired with post-mortem brain pathology**—a style of medicine that dominated neurology and psychiatry in the late 19th and early 20th centuries. The key idea was: *if you can link symptoms during life to consistent brain changes after death, you can define a disease.*

The first clear case (early 1900s)

- **1901–1906:** German physician **Alois Alzheimer** evaluated a patient (Auguste D.) with progressive memory loss, confusion, language problems, and behavioral changes at an unusually early age.
- **How he studied it:**
  i. detailed symptom tracking while the patient was alive, and
  ii. **microscopic examination of the brain after death** using histological stains (the essential "tool" of the era).
- **What he found:** the two hallmark lesions we still talk about:
  - **Amyloid plaques** (extracellular deposits)

- **Neurofibrillary tangles** (intracellular filamentous pathology)

When it became “Alzheimer’s disease”

- **1910:** Emil Kraepelin (a major figure in psychiatric classification) popularized the term **“Alzheimer’s disease”** in his textbook, helping turn a single striking case into a named diagnostic entity.

Why this matters: AD started as a *clinicopathological* definition: a recognizable clinical syndrome + characteristic brain lesions.

🧪 1900–2000: the most important research progress (selected milestones)

A useful way to see the century is in three waves:

1. defining the disease by pathology (early century),
2. recognizing it as the common cause of dementia in older adults (mid-century), and
3. uncovering molecular mechanisms and genetics (late century).

🧱 1900–1950: defining the pathology, slowly building consensus

This period was less about “big molecular breakthroughs” and more about **confirming and standardizing what AD is**.

Key developments

- **1906–1910:** Plaques and tangles described; AD named and placed into medical classification.
- **1920s–1950s:** Improved staining and microscopy methods gradually made plaques/tangles easier to visualize and compare across patients.

Why it matters: it cemented the idea that AD is a **distinct brain disease**, not just “normal aging.”

🧠 1950–1975: Alzheimer's becomes central to dementia research

Mid-century work reframed AD from a rare early-onset curiosity into a major public health problem.

Most important shifts

- **AD recognized as the leading pathology behind many cases of late-life dementia.**
  Earlier, "senile dementia" was often treated as a separate, expected part of aging. Research increasingly supported that many "senile" cases shared the same plaques/tangles seen in Alzheimer's original descriptions.

The cholinergic hypothesis emerges (1970s)

- Researchers found major loss of **cholinergic function** (especially neurons linked to memory circuitry) in AD.
- This led to the **cholinergic hypothesis**: reduced acetylcholine signaling contributes to cognitive symptoms.

Why it matters: it provided the first strong, actionable neurochemical target for symptomatic treatment.

🧬 1975–2000: molecular biology and genetics transform the field

This is where AD research accelerates—moving from describing lesions to identifying their **molecular components**, then to **genes and mechanisms**.

Identifying what plaques and tangles are made of

- **1980s:** The major component of plaques was identified as **amyloid-β (Aβ)**, derived from a larger protein precursor (**APP**, amyloid precursor protein).

- **1980s–1990s:** Tangles were linked to abnormal accumulation of **tau**, a microtubule-associated protein, in an altered (often hyperphosphorylated) form.

Why it matters: plaques and tangles became not just "things you see," but **biochemical targets** you can measure, model, and try to modify.

Genetics: separating risk from causation

Some discoveries explained *why some families get AD early*; others explained *why risk rises in the general population*.

- **Early-onset familial AD (rare but decisive):**
  - **APP mutations** (identified in the early 1990s) showed that altering APP/Aβ biology can cause AD.
  - **Presenilin 1 and Presenilin 2 (PSEN1/PSEN2)** mutations (mid-1990s) were found in families with early-onset AD; these genes are central to the enzymatic processing that generates Aβ species.
  - These findings strongly supported a mechanistic "amyloid pathway" in at least some forms of AD.
- **Late-onset AD risk (common form):**
  - **APOE ε4** (early 1990s) was identified as a major genetic risk factor for late-onset AD.
  - Important nuance: APOE ε4 **raises risk** and tends to **lower age of onset**, but it is not deterministic.

Why it matters: genetics anchored AD in **testable causal biology** (familial genes) while also explaining population risk patterns (APOE).

Diagnostic framework becomes formalized (1980s)

- **1984:** Widely adopted clinical criteria (NINCDS-ADRDA era) standardized diagnosis into categories like "probable" and "possible" AD during life, with confirmation by pathology after death.

Why it matters: standard criteria enabled large clinical studies and comparable research cohorts—crucial for trials and epidemiology.

Treatments: first symptomatic drugs (1990s)

- Building on the cholinergic hypothesis:
  - **Cholinesterase inhibitors** were developed and approved in the 1990s (symptomatic benefit for cognition/function in some patients).
- These did **not** stop disease progression, but they established AD as treatable (at least symptomatically) and made clinical trials a central research engine.

Why it matters: it shifted AD from "diagnose only" to "diagnose and manage," and created infrastructure for therapeutic research.

📅 Compact timeline of major milestones (1900–2000)

Below is a selective timeline of "cornerstone" findings (not every discovery, just the ones that redirected the field).

Era	Most important finding	Why it changed the field
1906–1910	Alzheimer describes plaques & tangles; disease is named	Defines AD as a clinicopathological entity
Mid-1900s	"Senile dementia" increasingly linked to AD pathology	AD becomes the central explanation for common dementia
1970s	Cholinergic deficits identified	First strong neurochemical target; drives symptomatic therapies

Era	Most important finding	Why it changed the field
1980s	Aβ identified as core plaque component	Launches modern amyloid biology and biomarker thinking
Late 1980s–1990s	Tau identified as key tangle component	Establishes a second major molecular pathway (tau pathology)
Early–mid 1990s	APOE ε4 risk factor discovered	Explains major population risk; genetics enters mainstream AD research
Early–mid 1990s	APP, PSEN1, PSEN2 mutations discovered	Demonstrates causal molecular mechanisms in familial AD
1990s	Cholinesterase inhibitors introduced	First broadly used symptomatic treatments; scales clinical trials

💡 Takeaways (what the 1900–2000 story really says)

- AD research began with **microscope-based pathology linked to clinical symptoms**—a powerful approach that still underpins diagnosis today.
- The mid-century pivot was recognizing AD as the **dominant pathology behind late-life dementia**, turning it into a major research priority.
- Late-century breakthroughs identified **Aβ and tau** and revealed **key genes** (APP/PSEN1/PSEN2, APOE), transforming AD into a mechanistic disease with targets—while also making clear that symptom treatments (like cholinesterase [illegible] progression.

What is plaque and tangle in respect?

👀 **Monica**

Plaques and tangles are the two signature brain lesions historically used to define Alzheimer's disease. They're different in **where they form**, **what they're made of**, and **what they tend to do to neurons**. Here's the clean distinction.

🧩 Plaques (amyloid plaques): "outside the neuron"

Plaques are **clumps of protein fragments that accumulate in the space between brain cells** (the extracellular space).

What they're made of

- Mostly **amyloid-β (Aβ)** peptides.
- Aβ is produced when a larger membrane protein, **APP (amyloid precursor protein)**, is cut by specific enzymes (often described as "secretases").

Where they are

- **Outside neurons**, in the tissue between cells.
- Commonly found in cortex and hippocampus-related regions involved in memory.

Why they matter

- Plaques (and especially smaller Aβ assemblies *before* plaques fully form, called **soluble Aβ oligomers**) are linked to:
  - disrupted synaptic signaling (communication between neurons)
  - inflammation and activation of immune cells in the brain (microglia/astrocytes)
  - downstream changes that can promote neuronal stress and degeneration

Nuance: Many researchers think **soluble Aβ forms** can be more directly toxic to synapses than the big “mature” plaque deposits themselves.

🧵 Tangles (neurofibrillary tangles): “inside the neuron”

Tangles are **twisted fibers that build up inside neurons** (intracellular pathology).

What they’re made of

- Mostly **tau** protein.
- Tau normally stabilizes **microtubules**—the cell’s internal “tracks” that move nutrients and other cargo along the neuron.

What goes wrong

- Tau becomes chemically altered (often discussed as **hyperphosphorylated**), detaches from microtubules, and misfolds/aggregates into filamentous structures.

Where they are

- **Inside neurons**, especially in regions crucial for memory and navigation (e.g., hippocampus and connected cortex), and they spread in a characteristic pattern as disease advances.

Why they matter

- Tangles correlate strongly with:
  - neuron dysfunction and death
  - severity of cognitive impairment
- Loss of tau’s normal function + gain of toxic aggregation both contribute:
  - disrupted transport inside neurons
  - structural collapse of neuronal processes

🔍 Quick side-by-side (the “in respect” comparison)

- **Location**
  - Plaques: **extracellular** (between cells)
  - Tangles: **intracellular** (within neurons)
- **Main protein**
  - Plaques: **amyloid-β**
  - Tangles: **tau**
- **Common interpretation**
  - Plaques/Aβ: often viewed as an **upstream trigger** (especially in many genetic forms of AD)
  - Tangles/tau: often track **neuronal injury and symptom severity** more closely

🧠 Big-picture relationship (how they’re often linked)

A widely taught model is:

1. abnormal Aβ production/clearance → Aβ accumulation (oligomers → plaques)
2. this promotes tau changes → tangles
3. tangles + synapse loss → cognitive decline

This is a simplified map—real brains are messier—but it captures why both lesions are central in Alzheimer’s research.

> Could you further introduce the research progress of AD after 2000 till now 2025?

👀 **Monica**

After 2000, Alzheimer's research shifts from "what does the brain look like at autopsy?" to "can we **detect**, **stage**, and **treat** the biology in living people?"—with big leaps in biomarkers (PET, CSF, blood), genetics/immune biology, and finally the first wave of disease-modifying therapies that target amyloid. Below is a curated, **most-important-findings** view from **2000–2025**.

🧭 2000–2025 at a glance: what changed?

Three breakthroughs define this era:

1. **Biomarkers made AD measurable during life** (amyloid PET, tau PET, CSF, then blood tests).
2. **Genetics + immunology reframed AD as involving microglia/innate immunity and lipid biology**, not only neurons.
3. **Anti-amyloid antibodies finally showed clinical benefit in early AD**, alongside new safety/implementation issues (ARIA, monitoring, selection).

🔬 2000–2010: the "living biology" era begins

This decade laid the foundation for staging AD before dementia.

Biomarkers become real clinical-research tools

- **CSF biomarkers mature** (widely replicated):
  - Lower **Aβ42** (reflecting brain amyloid deposition)
  - Higher **total tau** and **phosphorylated tau** (reflecting neuronal injury/tau pathology)
- **Amyloid PET imaging emerges (mid-2000s)**, allowing researchers to see amyloid deposition in living humans and to discover that:

- Amyloid can accumulate **years (often decades)** before dementia symptoms.
- Some cognitively normal older adults are amyloid-positive, shaping the idea of a **preclinical** stage.

Clinical trial infrastructure grows (and sobers expectations)

- Many disease-modifying candidates fail in trials, helping the field realize:
  - Treating too late may be a core reason for failure.
  - Better **patient selection** (biomarker-confirmed AD) is essential.

Treatment milestone (symptomatic)

- **Memantine** (NMDA receptor antagonist) becomes a standard symptomatic option for moderate–severe AD (often alongside cholinesterase inhibitors).
  Important, but not disease-modifying.

🧬 2010–2019: genetics, immunity, and a new definition of AD

This decade is about *what causes risk* and *how to define AD biologically*.

Genetics highlights immune and lipid pathways

Large genetic studies (GWAS) and sequencing expand risk genes beyond APOE, pointing strongly to:

- **Microglial/innate immune function** (a landmark example: **TREM2** variants associated with substantially increased AD risk)
- **Lipid handling and endosomal/lysosomal pathways**

Why it matters: AD is increasingly seen as involving a network: amyloid/tau biology plus **immune response, lipid transport, cellular**

clearance, and vascular/metabolic factors.

Tau PET and “tau tracks symptoms”

- **Tau PET imaging** arrives (mid-to-late 2010s), showing in living humans that:
  - Tau distribution aligns closely with **neurodegeneration and cognitive impairment severity**.
  - Amyloid is often widespread earlier; tau spread relates more to clinical decline.

New research frameworks redefine AD

- The field moves toward defining AD by **biology**, not symptoms alone:
  - **Preclinical AD** and **MCI due to AD** concepts expand.
  - The **A/T/N framework (2018)** organizes biomarkers into:
    - **A** (amyloid)
    - **T** (tau)
    - **N** (neurodegeneration/neuronal injury)

Prevention science gets serious

- Multidomain lifestyle trials (notably **FINGER**, 2015) support that managing vascular/metabolic risk, exercise, cognitive engagement, and diet can help maintain cognition in at-risk older adults.
- Major consensus efforts quantify **modifiable risk factors** (hearing loss, hypertension, smoking, diabetes, inactivity, etc.), pushing AD prevention into mainstream public health thinking.

💊 2020–2025: disease-modifying therapy arrives (with caveats), and blood tests take off

This period is the most clinically disruptive since the disease was first described.

Anti-amyloid monoclonal antibodies show clinical benefit in early AD

Key therapeutic shift: **removing brain amyloid can slow decline modestly** in selected patients with early symptomatic AD (MCI or mild dementia) who are biomarker-confirmed amyloid-positive.

Milestones (high level):

- **Aducanumab** (FDA accelerated approval, 2021): historic and controversial; highlighted regulatory and evidence debates.
- **Lecanemab** (FDA traditional approval, 2023 in the US): clearer evidence of clinical slowing in early AD; broad impact on care pathways (infusion centers, monitoring).
- **Donanemab** (FDA approval, 2024 in the US): another amyloid-targeting antibody with a treatment strategy often framed around reaching amyloid clearance targets.

Core limitation: benefits are **real but modest**, and these therapies are not cures.

Safety + implementation become central (ARIA)

A major “new” issue for the field is that effective amyloid removal can cause brain imaging abnormalities:

- **ARIA-E** (edema/swelling) and **ARIA-H** (microhemorrhages/siderosis)
- Risk is higher in some groups (notably **APOE ε4 carriers**) and requires MRI monitoring and careful selection.

This changes AD care into something closer to oncology-style risk/benefit planning.

Blood-based biomarkers transform diagnosis and trial screening

From ~2020 onward, blood tests rapidly improve, especially:

- **Plasma phospho-tau species** (e.g., p-tau181, p-tau217) tracking AD pathology with high accuracy in many studies
- Blood **Aβ42/40** ratios and related panels
- These enable:
  - cheaper, scalable screening
  - earlier identification of likely AD biology
  - much faster recruitment for prevention and early-intervention trials

Clinical reality: by 2025, blood biomarkers are increasingly used in specialized settings, often as part of a workflow that still confirms with PET/CSF depending on the context and regulation.

Better recognition of "mixed" dementias and look-alikes

Research emphasizes that many older adults have **multiple pathologies**:

- AD + vascular brain injury
- AD + Lewy body pathology
- AD + TDP-43–related changes (often discussed under frameworks like LATE) This improves diagnostic nuance and explains why "pure amyloid" approaches don't help everyone equally.

🗓️ Select timeline (2000–2025) — only the biggest shifts

Period	Most important progress	Why it mattered
2000–2006	CSF Aβ/tau biomarkers consolidate	Enables in-life biological evidence of AD
Mid-2000s	**Amyloid PET** becomes available	Shows amyloid accumulates long before dementia

Period	Most important progress	Why it mattered
2010–2015	GWAS/rare variants highlight immunity; **TREM2**	Reframes mechanisms toward microglia/clearance/lipid pathways
Mid–late 2010s	**Tau PET** + staging improvements	Tau burden maps more closely to symptoms
2018	**A/T/N** biological research framework	Standardizes AD as a biomarker-defined continuum
2021	Aducanumab accelerated approval (US)	Signals new era; sparks evidence/regulatory debate
2023–2024	Lecanemab then donanemab approvals (US)	Confirms disease-modifying benefit in early AD; forces new care models
2020–2025	Blood biomarkers (esp. plasma p-tau) mature	Makes large-scale screening and early detection practical

💡 Takeaways (what "progress" means by 2025)

- AD is now widely treated as a **continuum**: silent pathology → subtle symptoms → dementia.
- The field has moved from "AD = memory loss" to "AD = **amyloid + tau biology** (plus immune/lipid/vascular context)."
- **Disease-modifying therapy exists** (anti-amyloid antibodies), but:
  - works best **early**
  - requires **biomarker confirmation**
  - carries meaningful monitoring burdens and risks (ARIA)

- The next frontier is combining:
    - earlier identification (blood tests),